B6 Condensed-Matter Physics — Model Solutions, 2017

Question 1 — FCC lattice, X-ray diffraction in zincblende GaAs

Problem. Conventional FCC cube of side a. Describe the reciprocal lattice; find d_{111}, the angle between (111) and ($\bar{1}11$), and between (111) and (100). For zincblende GaAs ($a = 5.653 \times 10^{-10}$ m, basis Ga at 0, As at $\frac{1}{4}\frac{1}{4}\frac{1}{4}$), find scattering angles and relative intensities of the lowest six reflections at $E = 8\,\mathrm{keV}$. Compare with AlAs, $a = 5.660 \times 10^{-10}$ m.

Reciprocal lattice

Real-space FCC primitive vectors give a BCC reciprocal lattice with conventional cube side $4\pi/a$.

$$\boxed{\text{Reciprocal of FCC}(a) = \text{BCC with conventional cube } 4\pi/a.}$$

The first Brillouin zone is the truncated octahedron (Wigner–Seitz cell of BCC).

Spacing d_{111} and link to reciprocal lattice

Cubic spacing formula:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

Substitute $(hkl) = (111)$:

$$\boxed{d_{111} = \frac{a}{\sqrt{3}}.}$$

$\mathbf{G}_{111} = (2\pi/a)(1, 1, 1)$ points along the cube body-diagonal; its tip is the L-point on the BZ boundary.

Angle between (111) and ($\bar{1}11$) planes

Plane normals lie along $\mathbf{G}_{hkl} \propto (h, k, l)$. Angle between normals:

$$\cos\alpha = \frac{(1, 1, 1) \cdot (-1, 1, 1)}{|(1, 1, 1)|\,|(-1, 1, 1)|} = \frac{1}{3}.$$

$$\boxed{\alpha_{(111),(\bar{1}11)} = \arccos(1/3) \approx 70.53^\circ.}$$

Angle between (111) and (100) planes

$$\cos\beta = \frac{(1, 1, 1) \cdot (1, 0, 0)}{\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

$$\boxed{\beta_{(111),(100)} = \arccos\left(1/\sqrt{3}\right) \approx 54.74^\circ.}$$

Powder diffraction from zincblende GaAs

Wavelength from $E = hc/\lambda$:

$$\lambda = \frac{hc}{E} = \frac{(6.626\times10^{-34}\,\mathrm{J\,s})(3.0\times10^{8}\,\mathrm{m\,s^{-1}})}{(8000)(1.602\times10^{-19}\,\mathrm{J})}.$$

$$\lambda = 1.55\times10^{-10}\,\mathrm{m} = 1.55\,\text{Å}.$$

Bragg condition: $\lambda = 2d_{hkl}\sin\theta$, scattering angle 2θ.
Zincblende structure factor (FCC+basis Ga $\mathbf{0}$, As $\frac14\frac14\frac14$):

$$F_{hkl} = F_{\mathrm{FCC}}\left[f_{\mathrm{Ga}} + f_{\mathrm{As}}\,e^{i\pi(h+k+l)/2}\right].$$

$F_{\mathrm{FCC}} = 4$ if h,k,l all even or all odd; zero otherwise.
Three FCC-allowed classes:

- all odd: $|F|^2 \propto f_{\mathrm{Ga}}^2 + f_{\mathrm{As}}^2$ (medium).
- all even, $h+k+l = 4n$: $|F|^2 \propto (f_{\mathrm{Ga}} + f_{\mathrm{As}})^2$ (strong).
- all even, $h+k+l = 4n+2$: $|F|^2 \propto (f_{\mathrm{Ga}} - f_{\mathrm{As}})^2$ (weak; non-zero because $Z_{\mathrm{Ga}} = 31 \neq Z_{\mathrm{As}} = 33$).

Lowest six reflections by $N = h^2+k^2+l^2$, with $\sin\theta = \lambda\sqrt{N}/(2a)$:

$$\frac{\lambda}{2a} = \frac{1.55}{2\times5.653} = 0.1371.$$

(hkl)	N	$d_{hkl}/\text{Å}$	$\sin\theta$	θ	2θ	class
(111)	3	3.264	0.2374	13.74°	27.47°	odd, medium
(200)	4	2.827	0.2741	15.91°	31.82°	even $4n+2$, weak
(220)	8	1.999	0.3877	22.81°	45.63°	even $4n$, strong
(311)	11	1.704	0.4546	27.04°	54.08°	odd, medium
(222)	12	1.632	0.4748	28.35°	56.70°	even $4n+2$, weak
(400)	16	1.413	0.5483	33.25°	66.50°	even $4n$, strong

Estimate atomic form factors at $\sin\theta/\lambda$ small using $f \approx Z$:

$$(f_{\mathrm{Ga}} + f_{\mathrm{As}})^2 \approx 64^2 = 4096.$$

$$(f_{\mathrm{Ga}}^2 + f_{\mathrm{As}}^2) \approx 31^2 + 33^2 = 2050.$$

$$(f_{\mathrm{As}} - f_{\mathrm{Ga}})^2 \approx 2^2 = 4.$$

Multiplicities m_{hkl}: (111): 8, (200): 6, (220): 12, (311): 24, (222): 8, (400): 6.
Relative intensity $I \propto m_{hkl}|F_{hkl}|^2$ (Lorentz–polarisation/Debye–Waller suppressed):

$$I_{111} : I_{200} : I_{220} : I_{311} : I_{222} : I_{400}$$

$$\propto 8(2050) : 6(4) : 12(4096) : 24(2050) : 8(4) : 6(4096)$$

$$= 16400 : 24 : 49152 : 49200 : 32 : 24576.$$

Normalise to (220) = 100:

$$\boxed{I_{111}{:}I_{200}{:}I_{220}{:}I_{311}{:}I_{222}{:}I_{400} \approx 33{:}0.05{:}100{:}100{:}0.07{:}50.}$$

The (200) and (222) peaks are nearly extinct: they would vanish exactly if Ga and As were identical (as in the diamond structure of Si).

AlAs comparison: $\Delta(2\theta)$ for the (400) reflection

Differentiate Bragg's law at fixed λ:

$$\cos\theta\,\mathrm{d}\theta = -\frac{\lambda\sqrt{N}}{2}\frac{\mathrm{d}a}{a^2} = -\sin\theta\,\frac{\mathrm{d}a}{a}.$$

$$\mathrm{d}(2\theta) = -2\tan\theta\,\frac{\mathrm{d}a}{a}.$$

For (400) with $\theta = 33.25^\circ$ and $\Delta a/a = 7\times10^{-4}/5.653 = 1.24\times10^{-3}$:

$$\Delta(2\theta) = -2\tan(33.25^\circ)(1.24\times10^{-3}) = -1.62\times10^{-3}\,\text{rad}.$$

$$\boxed{\Delta(2\theta)_{(400)} \approx -0.093^\circ \quad \text{(AlAs at smaller } 2\theta\text{)}.}$$

This is at the edge of a standard powder diffractometer ($\sim 0.05^\circ$); higher-order reflections amplify $\tan\theta$ and so the splitting. Alternatively, exploit the Z-difference: $Z_{\text{Al}} = 13$ vs $Z_{\text{Ga}} = 31$ change $|F|^2$ for the all-odd class by a factor $(13^2+33^2)/(31^2+33^2) \approx 0.62$, and the weak even "$4n+2$" reflections (e.g. $(200), (222)$) become much stronger in AlAs because $|f_{\text{As}} - f_{\text{Al}}| = 20$ vs 2 for GaAs. *Resonant (anomalous) scattering* near a Ga or As K-edge is the cleanest discriminator.

Question 2 — Free-electron metal: heat capacity, entropy, Pauli χ, Na

Problem. From $U = \int_0^\infty E\,g(E)f(E)\,\mathrm{d}E$ explain g, f and their T-dependence; argue $C = \gamma T$ with $\gamma = \alpha k_B^2 g(E_F)$. Derive low-$T$ entropy and Pauli susceptibility. For Na (BCC, monovalent, $R_H = -25.0\times10^{-11}\,\text{m}^3\,\text{C}^{-1}$, $m^* = 1.3m_e$), find a, E_F, γ, χ_p.

Meaning of $g(E)$ and $f(E)$; rough $C = \gamma T$ argument

$g(E)\,\mathrm{d}E$: number of single-electron states per unit volume in $[E, E+\mathrm{d}E]$ (sums spin). Independent of T.
$f(E) = [\exp((E-\mu)/k_BT) + 1]^{-1}$: Fermi–Dirac occupation. Carries all T-dependence (also weakly through $\mu(T)$).

Only electrons within $\sim k_BT$ of E_F are thermally active.

$$\Delta N \sim g(E_F)\,k_BT.$$

Each gains energy $\sim k_BT$:

$$\Delta U \sim g(E_F)(k_BT)^2.$$

Differentiate w.r.t. T:

$$C = \frac{\partial U}{\partial T} \sim 2k_B^2 g(E_F)\,T.$$

Exact Sommerfeld coefficient is $\alpha = \pi^2/3$:

$$\boxed{C = \gamma T, \qquad \gamma = \tfrac{\pi^2}{3}k_B^2 g(E_F).}$$

Low-temperature entropy

Use $C = T(\partial S/\partial T)_V$:

$$\frac{\partial S}{\partial T} = \frac{C}{T} = \gamma.$$

Integrate from $T = 0$ (third law: $S(0) = 0$):

$$\boxed{S(T) = \gamma T.}$$

Pauli susceptibility

Apply $\mathbf{B} = B\hat{\mathbf{z}}$. Spin Zeeman energy $\mp\mu_B B$ shifts the spin-up/down half-bands:

$$g_\uparrow(E) = \tfrac{1}{2}g(E + \mu_B B), \quad g_\downarrow(E) = \tfrac{1}{2}g(E - \mu_B B).$$

Net magnetisation per volume:

$$M = \mu_B \int_0^\infty \big[g_\uparrow(E) - g_\downarrow(E)\big] f(E)\,\mathrm{d}E.$$

Linearise in B via $g(E \pm \mu_B B) \approx g(E) \pm \mu_B B\, g'(E)$:

$$M = \mu_B^2 B \int_0^\infty \big[-g'(E)\big] f(E)\,\mathrm{d}E.$$

Integrate by parts; at $T \to 0$, $-\partial f/\partial E = \delta(E - E_F)$:

$$M = \mu_B^2\, g(E_F)\, B.$$

Susceptibility $\chi_p = \mu_0 M/B$:

$$\boxed{\chi_p = \mu_0 \mu_B^2\, g(E_F).}$$

Sodium: lattice constant from R_H

Free-electron Hall coefficient:

$$R_H = -\frac{1}{ne}.$$

Solve for n:

$$n = -\frac{1}{R_H e} = \frac{1}{(25.0 \times 10^{-11})(1.602 \times 10^{-19})} = 2.50 \times 10^{28}\,\mathrm{m}^{-3}.$$

BCC monovalent: 2 atoms per conventional cell, each donates one electron, so $n = 2/a^3$:

$$a = \left(\frac{2}{n}\right)^{1/3} = \left(\frac{2}{2.50 \times 10^{28}}\right)^{1/3}.$$

$$\boxed{a_{\mathrm{Na}} = 4.31 \times 10^{-10}\,\mathrm{m} = 4.31\,\text{Å}.}$$

Sodium: E_F, γ, χ_p with $m^* = 1.3m_e$

Fermi wave-vector:

$$k_F = (3\pi^2 n)^{1/3} = (3\pi^2 \times 2.50 \times 10^{28})^{1/3} = 9.06 \times 10^9\,\mathrm{m}^{-1}.$$

Fermi energy:

$$E_F = \frac{\hbar^2 k_F^2}{2m^*} = \frac{(1.055 \times 10^{-34})^2 (9.06 \times 10^9)^2}{2(1.3)(9.11 \times 10^{-31})}.$$

$$E_F = 3.84 \times 10^{-19}\,\mathrm{J}.$$

$$\boxed{E_F \approx 2.40\,\mathrm{eV}.}$$

Density of states at E_F (free electron, per volume):

$$g(E_F) = \frac{3n}{2E_F} = \frac{3(2.50 \times 10^{28})}{2(3.84 \times 10^{-19})} = 9.76 \times 10^{46}\,\mathrm{J}^{-1}\,\mathrm{m}^{-3}.$$

Electronic specific-heat coefficient ():

$$\gamma_V = \tfrac{\pi^2}{3} k_B^2 g(E_F) = \tfrac{\pi^2}{3}(1.381 \times 10^{-23})^2 (9.76 \times 10^{46}).$$

$$\gamma_V = 61.2\,\mathrm{J\,K^{-2}\,m^{-3}}.$$

Convert to per mole using molar volume $V_m = N_A/n$:

$$\gamma = \gamma_V \frac{N_A}{n} = (61.2)\frac{6.022 \times 10^{23}}{2.50 \times 10^{28}}.$$

$$\boxed{\gamma_{\mathrm{Na}} \approx 1.48\,\mathrm{mJ\,mol^{-1}\,K^{-2}}.}$$

(Experimental: $\sim 1.4\,\mathrm{mJ\,mol^{-1}\,K^{-2}}$ – consistent.)

Pauli susceptibility ():

$$\chi_p = \mu_0 \mu_B^2 g(E_F) = (4\pi \times 10^{-7})(9.274 \times 10^{-24})^2 (9.76 \times 10^{46}).$$

$$\boxed{\chi_p \approx 1.05 \times 10^{-5} \quad \text{(SI, dimensionless).}}$$

Question 3 — Curie law, Curie–Weiss, mean-field ferromagnetism in Ni

Problem. Spin-1/2 paramagnet, n moments per volume in field B at T: derive M and $\chi = C/T$ with $C = n\mu_0\mu_B^2/k_B$. Add mean-field $B_{\mathrm{eff}} = \lambda\mu_0 M$: derive Curie–Weiss form $\chi = C/(T-\theta)$. For Ni ($\rho = 8908\,\mathrm{kg\,m^{-3}}$, $M_r = 58.693$, $T_C = 627\,\mathrm{K}$), find λ and B_{eff}.

(a) Curie law, spin-$1/2$ paramagnet

Two states $m_s = \pm 1/2$, energies $\mp \mu_B B$. Boltzmann weights $e^{\pm \mu_B B/k_B T}$:

$$\langle m_z \rangle = \mu_B \frac{e^{\mu_B B/k_B T} - e^{-\mu_B B/k_B T}}{e^{\mu_B B/k_B T} + e^{-\mu_B B/k_B T}} = \mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right).$$

Magnetisation per volume:

$$M = n\mu_B \tanh\left(\frac{\mu_B B}{k_B T}\right).$$

Linear regime $\mu_B B \ll k_B T$, $\tanh x \approx x$:

$$M = \frac{n\mu_B^2 B}{k_B T}.$$

Susceptibility $\chi = \mu_0 M/B$:

$$\boxed{\chi = \frac{C}{T}, \qquad C = \frac{n\mu_0\mu_B^2}{k_B}.}$$

(b) Mean-field Curie–Weiss

Replace external field by $B_{\text{tot}} = B + \lambda\mu_0 M$ in the linearised result:

$$M = \frac{C}{\mu_0 T}\,(B + \lambda\mu_0 M).$$

Collect M:

$$M\left(1 - \frac{\lambda C}{T}\right) = \frac{CB}{\mu_0 T}.$$

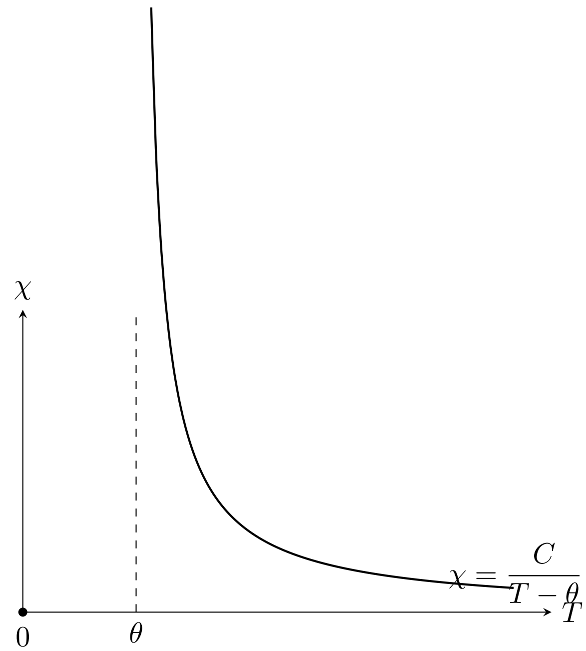
Solve for M:

$$M = \frac{CB}{\mu_0(T - \lambda C)}.$$

Susceptibility:

$$\boxed{\chi = \frac{C}{T - \theta}, \qquad \theta = \lambda C = \frac{\lambda n \mu_0 \mu_B^2}{k_B}.}$$

Significance: θ is the mean-field critical temperature (T_C). $\theta > 0$ favours ferromagnetic alignment; $\theta < 0$ antiferromagnetic. χ diverges as $T \to \theta^+$, signalling spontaneous order.



(c) Nickel: mean-field λ and B_{eff}

Number density of Ni atoms:

$$n = \frac{\rho N_A}{M_r} = \frac{(8908)(6.022 \times 10^{23})}{58.693 \times 10^{-3}} = 9.14 \times 10^{28}\,\text{m}^{-3}.$$

Curie constant from ():

$$C = \frac{n\mu_0\mu_B^2}{k_B} = \frac{(9.14 \times 10^{28})(4\pi \times 10^{-7})(9.274 \times 10^{-24})^2}{1.381 \times 10^{-23}}.$$

$$C = 0.715\,\text{K}.$$

Identify $\theta = T_C$ and use ():

$$\lambda = \frac{T_C}{C} = \frac{627}{0.715}.$$

$$\boxed{\lambda \approx 8.8 \times 10^2.}$$

Saturation magnetisation (spin-1/2, all moments aligned, low T):

$$M_{\text{sat}} = n\mu_B = (9.14 \times 10^{28})(9.274 \times 10^{-24}) = 8.48 \times 10^5\,\text{A}\,\text{m}^{-1}.$$

Mean-field exchange field:

$$B_{\text{eff}} = \lambda\mu_0 M_{\text{sat}} = (878)(4\pi \times 10^{-7})(8.48 \times 10^5).$$

$$\boxed{B_{\text{eff}} \approx 9.3 \times 10^2\,\text{T}.}$$

This is two orders of magnitude beyond the strongest laboratory magnets and far exceeds dipolar fields ($\mu_0\mu_B/r^3 \sim 0.1\,\text{T}$). B_{eff} cannot be a real magnetic field; it parametrises the *quantum exchange interaction*, which is electrostatic (Coulomb + Pauli) in origin.

(d) Why room-temperature Ni often shows $M = 0$

Below T_C a bulk specimen breaks into magnetic *domains* of opposite magnetisation, separated by Bloch walls. Domain formation lowers magnetostatic (stray-field) energy; the macroscopic vector sum of domain magnetisations averages to zero in an unmagnetised sample, even though each domain is locally saturated.

Question 4 — Intrinsic and doped semiconductors

Problem. Show $np = A(m_e^* m_h^*)^B T^C \exp(-E_g/k_BT)$, identify A, B, C. Find $\mu(T)$ for intrinsic case and the constant α. For Si ($E_g = 1.11\,\mathrm{eV}$, $m_e^* = 0.26m_e$, $m_h^* = 0.39m_e$, $\epsilon_r = 11.7$): donor binding energy of P; p at $T = 300\,\mathrm{K}$ for pure and $N_d = 10^{20}\,\mathrm{m}^{-3}$ doped Si.

(a) Carrier product np

Conduction-band density of states (parabolic, from E_c):

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(E - E_c)^{1/2}.$$

Boltzmann tail $f(E) \approx e^{-(E-\mu)/k_BT}$ valid when $E_c - \mu \gg k_BT$. Electron number density:

$$n = \int_{E_c}^{\infty} g_c(E)\, e^{-(E-\mu)/k_BT}\, \mathrm{d}E.$$

Substitute $x = (E - E_c)/k_BT$:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^* k_BT}{\hbar^2}\right)^{3/2} e^{-(E_c-\mu)/k_BT}\int_0^{\infty} x^{1/2}e^{-x}\mathrm{d}x.$$

Use $\int_0^\infty x^{1/2}e^{-x}\mathrm{d}x = \sqrt{\pi}/2$:

$$n = 2\left(\frac{m_e^* k_BT}{2\pi\hbar^2}\right)^{3/2} e^{-(E_c-\mu)/k_BT} \equiv N_c\, e^{-(E_c-\mu)/k_BT}.$$

Holes from $1 - f(E) \approx e^{-(\mu-E)/k_BT}$ in valence band, mirrored:

$$p = 2\left(\frac{m_h^* k_BT}{2\pi\hbar^2}\right)^{3/2} e^{-(\mu-E_v)/k_BT} \equiv N_v\, e^{-(\mu-E_v)/k_BT}.$$

Multiply () and (); μ cancels:

$$np = 4\left(\frac{k_BT}{2\pi\hbar^2}\right)^{3}(m_e^* m_h^*)^{3/2}\, e^{-E_g/k_BT}, \qquad E_g = E_c - E_v.$$

$$\boxed{np = A\,(m_e^* m_h^*)^B T^C\, e^{-E_g/k_BT}, \quad A = \frac{4k_B^3}{(2\pi\hbar^2)^3},\ B = \tfrac{3}{2},\ C = 3.}$$

(b) Intrinsic chemical potential $\mu(T)$

Charge neutrality (intrinsic): $n = p$. Equate () and ():

$$N_c\, e^{-(E_c-\mu)/k_BT} = N_v\, e^{-(\mu-E_v)/k_BT}.$$

Take logarithm:

$$-\frac{E_c-\mu}{k_BT} + \ln N_c = -\frac{\mu-E_v}{k_BT} + \ln N_v.$$

Solve for μ:

$$2\mu = E_c + E_v + k_BT\ln(N_v/N_c).$$

Substitute $N_v/N_c = (m_h^*/m_e^*)^{3/2}$:

$$\boxed{\mu(T) = \frac{E_c+E_v}{2} + \tfrac{3}{4}k_BT\ln\frac{m_h^*}{m_e^*}, \quad \alpha = \tfrac{3}{4}\ln(m_h^*/m_e^*).}$$

For Si: $\alpha = \frac{3}{4}\ln(0.39/0.26) = 0.304$.

(c) Phosphorus donor binding energy

Hydrogenic donor in dielectric medium with effective mass m_e^*:

$$E_R^* = R_H\,\frac{m_e^*/m_e}{\epsilon_r^2} = (13.6\,\mathrm{eV})\frac{0.26}{11.7^2}.$$

$$\boxed{E_R^* \approx 26\,\mathrm{meV}.}$$

At $T = 1000\,\mathrm{K}$, $k_BT \approx 86\,\mathrm{meV} \gg E_R^*$: donors are fully ionised (in fact already so above $\sim 50\,\mathrm{K}$).

(d) Hole density at $T = 300\,\mathrm{K}$: pure and P-doped Si

Effective densities of states at $T = 300\,\mathrm{K}$:

$$N_c = 2\left(\frac{(0.26)(9.109\times10^{-31})(1.381\times10^{-23})(300)}{2\pi(1.055\times10^{-34})^2}\right)^{3/2}.$$

$$N_c = 3.33\times10^{24}\,\mathrm{m}^{-3}.$$

$$N_v = N_c\left(\frac{0.39}{0.26}\right)^{3/2} = 6.11\times10^{24}\,\mathrm{m}^{-3}.$$

Intrinsic carrier density:

$$n_i^2 = N_cN_v\,e^{-E_g/k_BT}.$$

Evaluate exponent: $E_g/k_BT = 1.11/(2.585\times10^{-2}) = 42.94$:

$$e^{-42.94} = 2.25\times10^{-19}.$$

$$n_i^2 = (3.33\times10^{24})(6.11\times10^{24})(2.25\times10^{-19}) = 4.58\times10^{30}\,\mathrm{m}^{-6}.$$

$$n_i = 2.14\times10^{15}\,\mathrm{m}^{-3}.$$

(i) Pure Si: $n = p = n_i$:

$$\boxed{p_{\mathrm{intr}} \approx 2.1\times10^{15}\,\mathrm{m}^{-3}.}$$

μ sits near mid-gap; the small shift $\alpha k_B T = (0.304)(0.0259\,\mathrm{eV}) = 7.9\,\mathrm{meV}$ above mid-gap (because $m_h^* > m_e^*$).

(ii) n-type, $N_d = 10^{20}\,\mathrm{m}^{-3}$ (fully ionised, $N_d \gg n_i$): $n \approx N_d$ and $p = n_i^2/n$:

$$p = \frac{n_i^2}{N_d} = \frac{4.58 \times 10^{30}}{10^{20}}.$$

$$\boxed{p_{\text{n-type}} \approx 4.6 \times 10^{10}\,\mathrm{m}^{-3}.}$$

Position of μ: $\mu = E_c - k_B T \ln(N_c/N_d)$:

$$\mu = E_c - (0.0259\,\mathrm{eV}) \ln\!\left(3.33 \times 10^{24}/10^{20}\right) = E_c - 0.27\,\mathrm{eV}.$$

μ has shifted from near mid-gap up to $\sim 0.27\,\mathrm{eV}$ below E_c – well into the upper half of the gap, as required for an n-type material.